

Ahern Group Protocols

Imaging isolated and *in situ* bladder arterioles

- Bladders are removed from TRPV1-Cre:tomato mice, ventrally opened longitudinally from the bladder neck to the top of the dome.
- Under a dissecting microscope the urothelium is carefully peeled away using iris scissors to expose sub-urothelial arterioles.
- The preparation is placed in 500 μ L of physiological buffer containing 140 mM NaCl, 4 mM KCl, 1 mM $MgCl_2$, 1.2 mM $CaCl_2$, 10 mM HEPES, 5 mM glucose, pH=7.3, and anchored by platinum-wire tissue harps.
- If desired individual 3rd order arterioles (~20-40 μ m diameter) can be carefully microdissected using iris scissors. Isolated arterioles are placed in a micro-insert chamber (Ibidi Inc., volume 10 μ L) containing 5 μ L of physiological buffer.
- Stock solutions of capsaicin (1 M and 10 mM in EtOH) and the TRPV1 antagonist, BCTC (2 mM in EtOH) are prepared. From these stocks working solutions of 2 mM (*in situ* studies) and 10 μ M (isolated arterioles) capsaicin are prepared. A working solution containing both capsaicin (10 μ M) and BCTC (2 μ M) is also prepared.
- tdTomato is excited at 540 $\pm$ 12 nm and emitted fluorescence is passed through a 620 $\pm$ 30 nm bandpass filter. Experiments are digitally recorded at a frame rate of at least 1 per second.
- Isolated arteries are challenged with 5 μ L of 2x capsaicin (10 μ M) to generate a final concentration of 5 μ M capsaicin. Subsequently, 5 μ L is withdrawn from the chamber and replaced with 5 μ L of capsaicin/BCTC (10 μ M and 2 μ M) solution.
- For *in situ* preparations a higher capsaicin concentration (~1mM) is required due to the tissue absorption of the drug. 500 μ L of 2x (2 mM)

capsaicin is added to chamber to yield a final concentration of 1mM capsaicin.

- At the end of the experiment a solution containing KCl (50 mM) is used as a positive control.
- Arterial diameter is measured post-hoc at several locations along the artery using the edge-detection plug-in for ImageJ.